

9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation

Code:

```
clc;
clear;
close all;

%% Generate Input Signal
x = randn(1,1000);

%% Echo Signal
h = [0.8 0.5 0.3];
d = filter(h,1,x);

%% Parameters
mu = 0.01;      % LMS step size
mu_n = 0.5;     % NLMS step size
L = 3;          % Filter length
delta = 1e-6;   % Small constant

%% Initialize
w_lms = zeros(1,L);
w_nlms = zeros(1,L);

e_lms = zeros(1,length(x));
```

```
e_nlms = zeros(1,length(x));
```

```
%% LMS and NLMS
```

```
for n = L:length(x)
```

```
    % Input vector
```

```
    u = x(n:-1:n-L+1);
```

```
    % LMS
```

```
    y_lms = w_lms * u';
```

```
    e_lms(n) = d(n) - y_lms;
```

```
    w_lms = w_lms + mu * e_lms(n) * u;
```

```
    % NLMS
```

```
    y_nlms = w_nlms * u';
```

```
    e_nlms(n) = d(n) - y_nlms;
```

```
    w_nlms = w_nlms + ...
```

```
        (mu_n/(delta + u*u')) * e_nlms(n) * u;
```

```
end
```

```
%% MSE
```

```
mse_lms = movmean(e_lms.^2,50);
```

```
mse_nlms = movmean(e_nlms.^2,50);
```

```
%% Figure 1: Input and Echo
```

```
figure;  
plot(x);  
hold on;  
plot(d);  
grid on;  
legend('Input','Echo');  
title('Input and Echo Signal');  
xlabel('Samples');  
ylabel('Amplitude');
```

```
%% Figure 2: LMS Error
```

```
figure;  
plot(e_lms);  
grid on;  
title('LMS Output/Error');  
xlabel('Samples');  
ylabel('Error');
```

```
%% Figure 3: NLMS Error
```

```
figure;  
plot(e_nlms);  
grid on;  
title('NLMS Output/Error');  
xlabel('Samples');
```

```
ylabel('Error');
```

```
%% Figure 4: MSE Comparison
```

```
figure;
```

```
semilogy(mse_lms,'r','LineWidth',1.5);
```

```
hold on;
```

```
semilogy(mse_nlms,'b','LineWidth',1.5);
```

```
grid on;
```

```
legend('LMS','NLMS');
```

```
title('LMS vs NLMS MSE Comparison');
```

```
xlabel('Samples');
```

```
ylabel('MSE');
```

Output:





